

Physics 4700H: Assignment II

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Problem 1. Using notation appropriate to a continuous variable, show that the variance $\langle(\Delta x)^2\rangle = \langle x^2\rangle - \langle x\rangle^2$, where $\Delta x \equiv x - \langle x\rangle$.

Solution 1. *Proof.* In the continuous case

$$\begin{aligned}\langle x^2 \rangle &= \int x^2 p(x) dx, \\ \langle x \rangle^2 &= \left(\int x p(x) dx \right)^2, \\ \langle (\Delta x)^2 \rangle &= \int (\Delta x)^2 p(x) dx \\ &= \int (x - \langle x \rangle)^2 p(x) dx && \text{Definition} \\ &= \int (x^2 - 2x\langle x \rangle + \langle x \rangle^2) p(x) dx && \text{Binomial expansion} \\ &= \langle x^2 \rangle + \int \langle x \rangle^2 p(x) - 2x\langle x \rangle p(x) dx && \text{Definition of } \langle x^2 \rangle \\ &= \langle x^2 \rangle + \langle x \rangle^2 - 2\langle x \rangle \int x p(x) dx && \langle x \rangle \text{ is a constant, } \int p(x) dx = 1 \\ &= \langle x^2 \rangle + \langle x \rangle^2 - 2\langle x \rangle^2 && \text{Definition of } \langle x \rangle \\ &= \langle x^2 \rangle - \langle x \rangle^2 && \square\end{aligned}$$

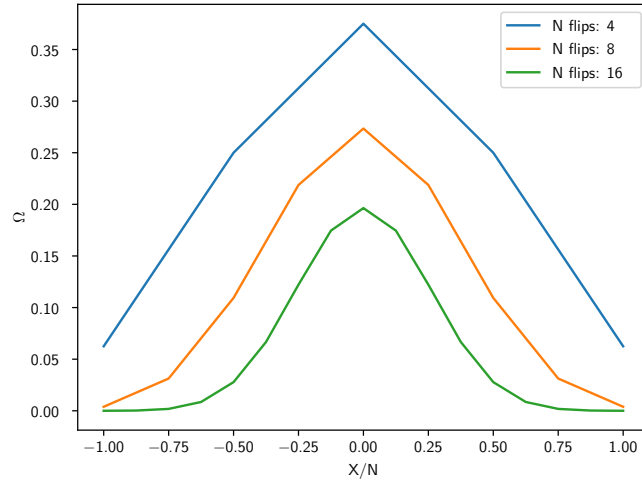
Problem 2. Consider the case of N tosses of a fair coin. The multiplicity function describes the number of ways a value of $X = N_H - N_T$ can be obtained,

$$\Omega(N, X) = \frac{N!}{\left[\frac{1}{2}(N+X)\right]! \left[\frac{1}{2}(N-X)\right]!}$$

- (a) Make a plot of $\Omega(N, X)/2^N$ versus X/N for three system sizes: $N = 4, 8, 16$. (Note that these distributions are discrete, not continuous.) Describe in words how the shape of the graph changes as N increases. You may construct the plot by hand (in which case show your calculations) or using a computer (in which case include your code).
- (b) Use the stronger form of Stirling's approximation to write an expression for $\ln \Omega(N, X)$ which is true when $N \gg 1$ and $|X|/N \ll 1$ (so $N \pm X$ also $\gg 1$).
- (c) Calculate the Taylor expansion of $\ln(1+x)$ for small x to second order.
- (d) Use your results in (b) and (c) to show that $\ln \Omega \approx N \ln(2 - X^2/(2N))$.
- (e) Using (d), find an approximate expression for Ω .

Solution 2.

- (a) The tails of the distribution flatten out as more probability “moves” towards the center and the lineshape approaches a Gaussian.



(b) The strong form of Stirling's approximation is, with the $\ln$

$$\ln(N!) \approx \left(N + \frac{1}{2}\right) \ln(N) - N + \frac{1}{2} \ln(2\pi)$$

so for our expression,

$$\begin{aligned}
\ln(\Omega(N, X)) &= \ln\left(\frac{N!}{\left[\frac{1}{2}(N+X)\right]! \left[\frac{1}{2}(N-X)\right]!}\right) \\
&= \ln(N!) - \ln\left(\left[\frac{1}{2}(N+X)\right]! \left[\frac{1}{2}(N-X)\right]!\right) \\
&\approx \left(N + \frac{1}{2}\right) \ln(N) - N + \frac{1}{2} \ln(2\pi) - \ln\left(\left[\frac{1}{2}(N+X)\right]!\right) + \ln\left(\left[\frac{1}{2}(N-X)\right]!\right) \\
&= \left(N + \frac{1}{2}\right) \ln(N) - N + \frac{1}{2} \ln(2\pi) - \ln\left(\left[\frac{1}{2}(N+X)\right]!\right) - \ln\left(\left[\frac{1}{2}(N-X)\right]!\right) \\
&= \left(N + \frac{1}{2}\right) \ln(N) - N + \frac{1}{2} \ln(2\pi) \\
&\quad - \left(\frac{1}{2}(N+X) + \frac{1}{2}\right) \ln\left(\frac{1}{2}(N+X)\right) - \frac{1}{2}(N+X) + \frac{1}{2} \ln(2\pi) \\
&\quad - \left(\frac{1}{2}(N-X) + \frac{1}{2}\right) \ln\left(\frac{1}{2}(N-X)\right) + \frac{1}{2}(N-X) - \frac{1}{2} \ln(2\pi) \\
&= \left(N + \frac{1}{2}\right) \ln(N) - N + \frac{1}{2} \ln(2\pi) \\
&\quad - \frac{1}{2}(N+X+1) \ln\left(\frac{1}{2}(N+X)\right) - X \\
&\quad - \frac{1}{2}(N-X+1) \ln\left(\frac{1}{2}(N-X)\right) \\
&\approx N \ln(N) - N - \frac{1}{2}N \ln\left(\frac{1}{2}(N+X)\right) - \frac{1}{2}N \ln\left(\frac{1}{2}(N-X)\right) \\
&= N \ln(N) - N - \frac{1}{2}N \left[\ln\left(\frac{1}{4}(N+X)(N-X)\right) \right] \\
&\approx N \left(\ln(N) - \frac{1}{2} \left[\ln\left(\frac{1}{4}(N^2 - X^2)\right) \right] \right) \\
&= N \ln\left(\frac{1}{2} \sqrt{\frac{N^2}{N^2 - X^2}}\right) \\
&= -\frac{N}{2} \ln\left(4 - 4\left(\frac{X}{N}\right)^2\right)
\end{aligned}$$

(c) For $f(x) = \ln(1+x)$, $f'(x) = \frac{1}{1+x}$, $f''(x) = -\frac{1}{(1+x)^2}$, $f'''(x) = \frac{2}{(1+x)^3}$. Then, about zero,

$$f(x) \approx x - \frac{x^2}{2} + \frac{x^3}{3}.$$

(d) Use your results in (b) and (c) to show that $\ln \Omega \approx N \ln(2 - X^2/(2N))$.

(e) Using (d), find an approximate expression for Ω .